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Data Mining for Discovering Consumer Behavioral Patterns in E-Commerce to Inform Marketing Strategies

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Abstract

The rapid growth of e-commerce has generated massive amounts of transaction data; however, high shopping cart abandonment rates remain a major challenge for the industry. This study applies data mining methods based on the CRISP-DM framework, with the aim of designing targeted marketing initiatives and identifying consumer behavior patterns. Following an evaluation of the K-Means, Gaussian Mixture Model (GMM), and Hierarchical Clustering models, the final results from Hierarchical Clustering with five clusters (K = 5) emerged as the optimal model for this study, as evidenced by the most effective balance between internal consistency and clear segment separation. Additionally, the process of detecting product associations to support marketing strategies utilizes the Association Rule Mining method via the Apriori algorithm. The research results are presented in an interactive dashboard that serves operational frameworks, such as product bundling packages aimed at reducing cart abandonment and consistently increasing sales conversion, as well as platform management workflows and strategic practitioners, to support the implementation of personalized promotions.

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Keywords: Data Mining;E-commerce;Shopping Cart Abandonment;K-Means Clustering;Apriori Algorithm;CRISP-DM.

1. Introduction

The digital technology has changed the way businesses work and made e-commerce a part of marketing plans . This change has also affected how people behave when they buy things online like how they search for information[1] and make purchasing decisions [2]. When people interact with e-commerce websites, they generate a lot of data that businesses can use to understand what customers like [3], [4]. Data from browsing activities and purchase histories

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gives us an idea of how people use e-commerce platforms [5], [6]. This helps organizations predict what customers will do in the future and make decisions [7].

However, a lot of people still leave their shopping carts without checking out which costs e-commerce businesses a lot of money [8]. In Indonesia things like how hard it's to check out and how much people trust the website affect whether they buy something [9], [10]. These problems show that traditional marketing methods are not working well. So, making decisions based on data has become very important. Data mining helps organizations find patterns in datasets and understand customer behavior better [11]. The CRISP-DM framework is a method that helps turn data into useful information [12].

One way to understand customers is to group them based on how they behave. This is called customer segmentation. Techniques like K-Means clustering help us find groups of customers that're similar [13], [14]. We can also look at which products people buy together which helps us recommend things to customers and sell more [15]. The Apriori algorithm is used a lot for finding associations and making rules that make sense [16]. To make sure we are grouping customers correctly we can use methods like the Elbow Method [17]. Previous studies have shown that combining customer segmentation and looking at which products people buy together helps us understand customer behavior better and improves business performance [18], [19], [20].

This study uses the CRISP-DM framework to analyze customer behavior in e-commerce [21], [22]. We are also using a dashboard to help us visualize and interpret the results, which will help us make better decisions [23]. By looking at customer segments, product associations and using the dashboard we hope to find insights that will help us make personalized marketing plans reduce shopping cart abandonment and improve the number of people who complete their purchases [24]. This study is about e-commerce and customer behavior. We are looking at customer segments and product associations to find insights. The digital technology and e-commerce are changing the way businesses work. We need to understand customer behavior to make marketing plans. The CRISP-DM framework and data mining are tools, for analyzing data and making decisions.

2. Literature Review

Understanding what consumers do when they shop online is not about looking at numbers it is also about knowing how people think and feel. When people shop online, they do not always make decisions. Sometimes they feel anxious or stressed. This can make them do things without thinking like adding things to their shopping cart without really wanting to buy them [25]. There are also things that can stop people from buying things online like worrying about if the site is safe or what happens if they want to return something [26]. These things can make people hesitant. They might leave the site without buying anything. All of this shows that understanding what consumers do online is complicated and involves different things.

To deal with this complexity companies are using data to make decisions. They use something called Big Data Analytics to look at how they are doing and to find useful information in large amounts of data [27]. They also use Machine Learning to predict what might happen in the future and to give advice on what to do. This helps companies find patterns in what consumers do that they might not have noticed otherwise [28]. From a point of view one-way companies understand their customers is by grouping them based on what they buy and how they behave. They use techniques like K-Means to do this, which commonly applied to group customers based on similarities in their transaction history and behavioral patterns [29]. This helps companies make marketing plans that are aimed at groups of customers. But it is not about using the right technique it is also about how you do the analysis.

To make sure the analysis is done correctly companies use frameworks like CRISP-DM. This framework gives a step-by-step guide on how to do the analysis from understanding the data to using the results. This is especially important when combining techniques like grouping customers and finding patterns in what they buy [30]. Without a framework the results might not be useful. In addition to analyzing data, it is also important to be able to understand and use the results. Business Intelligence that uses Artificial Intelligence can help companies make decisions by giving them more information [31]. Tools like dashboards can help non-technical people understand complex data and see patterns more easily [32].

It is also important to be able to adapt to changing conditions. Analyzing data in time and constantly monitoring it is crucial for keeping the analysis relevant. Companies that can do this are more likely to stay of their competitors [33]. Based on what has been discussed it is clear that understanding consumer behavior online requires a approach

that combines understanding how people think and behave using data mining techniques and using tools. While previous studies have looked at these things separately there is still a need for a approach that connects grouping customers with analyzing products in a framework. This is what this study aims to do by combining grouping customers and finding patterns in what they buy to get useful insights into consumer behavior in e-commerce.

3. Research Methodology

The methodology for analyzing consumer behavior based on data from transactions in e-commerce will be discussed in this chapter. This research employs data mining as its technique of choice to guarantee that the analysis conducted is rigorous, repeatable, and accurate. Segmentation along with market basket analysis will provide insights not only at a macro level but also a micro level. The overall sequence of this structured data pipeline, from data collection to dashboard visualization, is illustrated in Fig. 1.

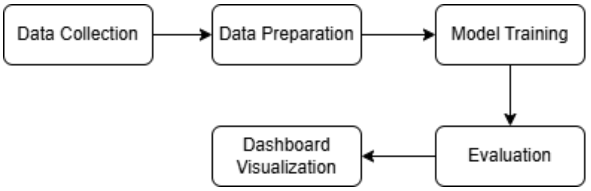


Fig. 1. Research Methodology

A. Research Methodology and Framework

This research applies an organized data mining methodology based on common analytical procedures applied during data science research. Structured procedure is required in order to obtain insightful knowledge from data by turning raw transactional data into something valuable and useful [34].

Furthermore, data analytics procedures need to pay attention not only to the technological aspect, but also to the generation of meaningful knowledge derived from data. It is important to ensure that a clear methodology will lead to generation of insight at every step of analysis process [35]. This research will be conducted using a structured data pipeline including the following steps: data collection, data preparation, modeling, evaluation, and interpretation.

B. Data Source and Analysis Units

The data set that has been chosen for this research project includes transaction logs of an e-commerce store, which include attributes like transactions ID, product ID, customer ID, quantity, and time of transactions. These attributes offer an all-encompassing view of consumer behavior on the platform. In order to observe different facets of consumer behavior, the data analysis is done on two distinct levels:

- Customer level: Aggregation of data to analyze customer behavior through parameters like purchase behavior, spending, number of purchases, and so forth. Clustering technique will be applied on this level.
- Transactions level: Treat each transaction as a basket of goods purchased in that transaction. Association rules are formed based on this level of data aggregation.

Data cleaning process is essential for ensuring that the data set is consistent and clean from errors.

C. Preparing and Preprocessing Data

Prior to implementing data mining processes, the data set is subjected to various preprocessing activities that aim at enhancing the quality of the data set and making it amenable to analysis. Considering the nature of transactional data, data preprocessing becomes very important. These activities entail:

- Data Cleaning: Elimination of entries that are invalid, missing, and duplicates
- Data Transformation: Transformation of data from its initial form into usable formats
- Feature Creation: Development of new features such as frequency of purchases and total spend
- Data Normalization: Scaling of numerical values to prevent magnitude differences from affecting clustering outcomes

Such preprocessing makes the data amenable to analysis using data mining methods.

D. Data Mining Techniques

Two types of data analysis techniques used for this research include customer segmentation and association rules

mining. The combination of the two contributes to the better understanding of consumers' behavior.

a. Customer Segmentation via Clustering

For the purpose of customer segmentation, clustering algorithms will be used to form groups according to similar user characteristics. K-means clustering is chosen as a clustering algorithm as it demonstrates high efficiency and is widely used for this kind of problem. The effectiveness of clustering techniques in relation to the task at hand is proved by the findings of previous researchers [36]. As a matter of fact, in addition to K-means algorithm, other types of clustering algorithms such as hierarchical clustering should be taken into account for the purpose of adding robustness. Number of clusters can be found based on Elbow method, Silhouette score, Davies-Bouldin index

b. Association Rule Mining for Market Basket Analysis

In order to examine relations between products, this research uses the approach of association rule mining based on Apriori algorithm. This approach allows discovering frequent itemsets and creating rules explaining relations between products within transactions. The technique of association rule mining is frequently applied in e-commerce for discovering co-purchase patterns and thus plays a key role in developing recommendation systems [37]. Moreover, the capacity to detect valuable patterns within transactional data may significantly improve decision-making and lead to better business performance [38]. Each rule obtained through analysis is considered by means of support – frequency, confidence – probability, lift – strength of association. Rules that fail to meet certain thresholds are excluded from further analysis.

E. Model Evaluation and Parameter Setting

In order to make sure that the results are valid and reliable, evaluations of both clustering and association rules output will be conducted. For clustering analysis, internal metrics like Silhouette Score and Davies-Bouldin index will be used for measuring the quality of clustering and Cluster interpretability is another important aspect that will be taken into account. For association rules: will be filtered on the basis of support, confidence, and lift. Also, irrelevant and low-quality rules will be omitted from the analysis.

In addition to evaluating models separately, an integrated approach will be applied in this study in the sense that association rule mining will be performed in clusters. Such an approach allows segment-specific patterns to be found. According to prior research, using clustering and association rules mining together enhances pattern discovery significantly [39].

F. The development, analysis, application, and reporting of insights

Finally, this phase of the methodology is concerned with the interpretation of analysis results and turning them into insightful conclusions. By applying the techniques of clustering and association rules together, one can achieve greater insight into the nature of customers and their interactions with products. Insights developed through this research will be used to recognize different customer groups, discover the patterns of their behavior within each group, and formulate strategies of targeted marketing. By making the connection between the analysis outcomes and their implementation, this paper guarantees relevance to practice.

4. Result and Discussion

4.1 Dataset Overview

The dataset used in this study consists of 25.000 interaction records collected from a group of 100 unique users. An initial assessment of the distribution reveals that events exhibit a clear funnel-shaped behavioral pattern: There is a significant drop of approximately 12% in the act of adding to cart, yet about 80% of all interactions are directed toward viewing products, ultimately resulting in a conversion rate of 8% for completed purchases. This hierarchical distribution underscores the inherent reduction pattern in the digital retail environment, where extensive browsing is far more common than the final commitment to make a transaction.

Table 1. Clustering Evaluation Summary

Model	K	Silhouette	DBI / CH
K-Means	2	0.880	DBI 0.339 | CH 134.34
GMM	5	0.814	DBI 0.373 | CH 281.24
Hierarchical	5	0.815	DBI 0.178 | CH 262.05

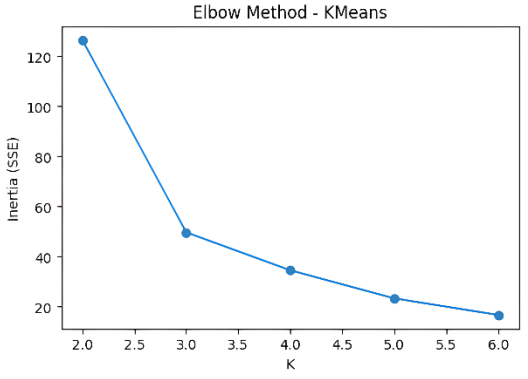


Fig. 2. Elbow Method for K-Means.

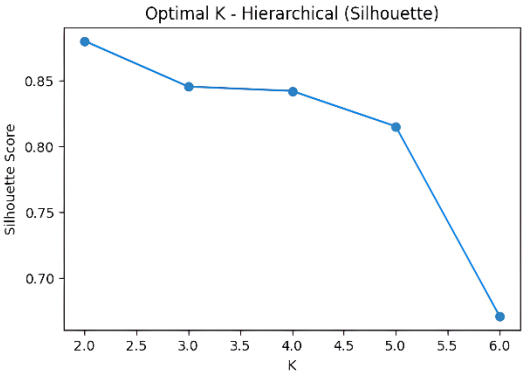


Fig. 3. Silhouette Score for Hierarchical Clustering

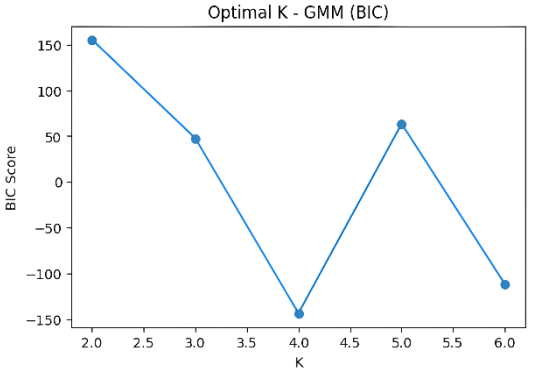


Fig. 4. Bayesian Information Criterion (BIC) Score for Gaussian Mixture Model.

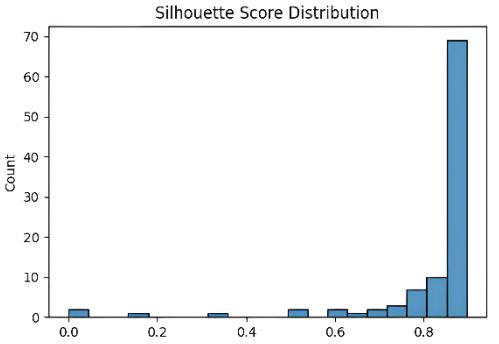


Fig. 5. Silhouette Score Distribution

The process of determining the optimal number of clusters using the inertia metric method is employed in the evaluation of the K-Means algorithm; this is because the algorithm operates by minimizing the sum of the squared distances between the data points and the centroids for each cluster. Inertia represents the degree of similarity within a formed cluster; thus, a lower value indicates that the data within a cluster are increasingly closer to one another. The evaluation shows a sharp decline in the elbow curve in the Sum of Squared Errors (SSE) between K = 2 and K = 3, with the rate of decline slowing down after that point, as illustrated in the elbow method plot for K-Means (Fig. 2). Adding clusters beyond that point yields results that tend to decline. The data indicates that adding clusters beyond that point yields results that tend to decline; however, this trend suggests that the most significant benefits in data organization occur in the early stages.

Meanwhile, the K-Means method offers a concept vastly different from Hierarchical Clustering; in Hierarchical Clustering, centroids are not used, which is evident in the inertia metric becoming less relevant for evaluation. Consequently, the research process employs alternative methods, such as the Silhouette Score, to measure the balance between data proximity within a cluster (cohesion) and the distance between clusters (separation). The resulting Silhouette Scores are concentrated between 0.8 and 0.9, indicating that most of the data is organized into compact and well-defined groups, which is further supported by the optimal K evaluation for Hierarchical clustering (Fig. 3). Although some lower outlier values were still found, their frequency is considered too low to compromise the structural integrity or overall reliability of the model, as clearly visible in the overall Silhouette score distribution histogram (Fig. 5).

As for the alternative approach, the Gaussian Mixture Model (GMM) was employed, a probabilistic method that model's data as a combination of Gaussian distributions. Consequently, the GMM is considered more focused on likelihood, which renders the use of metrics such as inertia or silhouette suboptimal. Instead, the Bayesian Information Criterion (BIC) method can be used as an alternative; this method balances model fit with data while penalizing model complexity. A lower BIC value indicates an optimal model, as it achieves a balance between model simplicity and accuracy. The visualization of these BIC scores across different numbers of clusters is depicted in (Fig. 4).

Although the elbow method can provide useful initial insights into K-Means performance, the final evaluation results of all models confirm that Hierarchical Clustering with five clusters (K=5) is the most superior option among the others. This specific configuration offers the most effective balance between internal consistency and clearer separation compared to other methods used in the study; consequently, Hierarchical Clustering with five clusters (K=5) was selected as the model for this research.

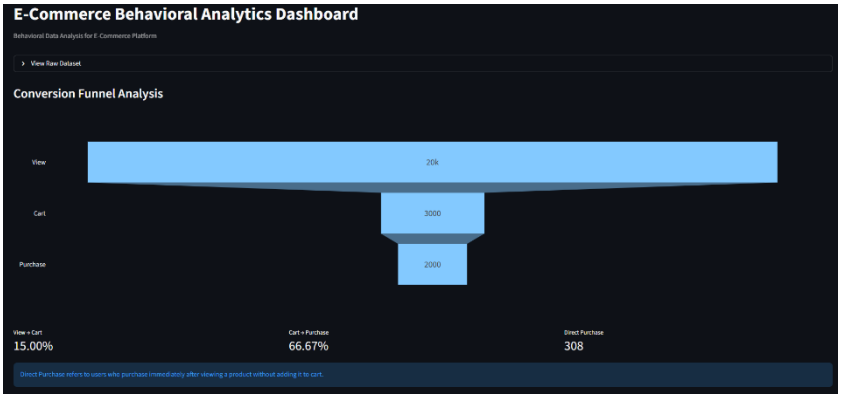


Fig. 6. Conversion Funnel Analysis

Fig. 6 depicts people's interaction with the website in processes such as when they view a product, add it to their cart and purchase it. Most individuals cease interacting with the website while viewing and adding products to their carts. This means that most individuals do not like the way the products are being presented to them. It may be that the products on the website are not being well represented or that it is hard to navigate the website. However, after adding an item to the cart people go ahead and make the purchase thus it shows that individuals are really keen on purchasing the products. From this, it can be concluded that it is essential to improve how things are done on the website since most individuals stop interacting with the website when they first view it. They should be encouraged to like the website.

4.2 Cluster Interpretation

Ultimately, the hierarchical solution yields several smaller, specialized segments alongside one large primary segment. This demonstrates that the existing dataset encompasses general customer behavior as well as several. However, it must be pointed out that both Cluster 3 and Cluster 4 consist of only one user each. Even though these two clusters have been included in the final clustering results based on Hierarchical Clustering owing to the exceptionally different behavioral pattern of users, they should be considered with caution within a marketing context. They are not the typical customer segments as they can be seen as behavioral outliers who display interaction in ways not typically displayed by other customers. Despite this, they are still worth considering in marketing as they show unique behavioral patterns that could be leveraged by a business in certain areas.

Table 2. Cluster Profile

Cluster	Profile summary	Observed pattern
0	93 users	Moderate browsing with balanced buying activity
1	2 users	Very high cart and purchase intensity
2	3 users	High browsing volume but no purchases
3	1 user	High carts and purchases in a compact segment
4	1 user	Extreme browsing activity with no purchases

The large clusters can be considered the primary audience for general campaigns, while the smaller clusters represent niche groups that may require more personalized interventions. Both can be viewed from a marketing perspective.

It is also important to note that data from users who do not make a purchase but are very close to the type of product they actually want requires several key points, such as better product information, a smoother payment process, and stronger offers.

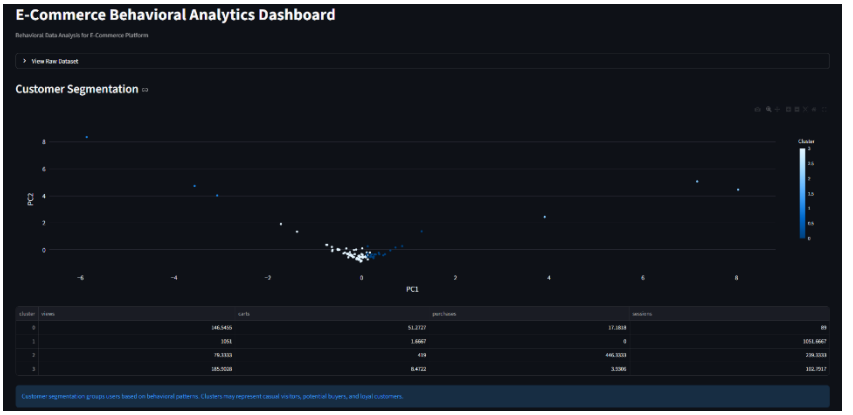


Fig. 7. Customer Segmentation

Results of the customer segmentation performed by clustering are illustrated in Fig. 7 in the form of a two-dimensional graph due to dimensionality reduction techniques. Each point on the graph is a particular user, while various colors denote clusters. Analysis of the distribution of the clusters shows diverse behavioral patterns among the customers, with clusters featuring intense user interactions, average engagement levels, and low conversion rates. Segmentation demonstrates that users do not exhibit identical behaviors and allows identifying various user segments. These segments can be used as a basis for developing marketing campaigns.

4.3 *Association Rule Mining*

Association rule mining identifies product relationships within consumers' shopping carts and uncovers behavioral patterns that are not immediately apparent through traditional sales analysis methods. Evaluation measures hold significant value for these relationships, with three core metrics used: Lift, which determines the strength of an association by comparing it to the probability that those items would be purchased independently by chance; Confidence, which measures the conditional probability or reliability of a purchase following an initial interaction; and finally, Support, which measures the joint frequency of a set of items within the entire dataset.

Table 3. Top Global Association Rule

Antecedent	Consequent	Support	Confidence	Lift
Artel (4201541)	LG (1801555)	0.033790	0.435294	3.782913
LG (1801555)	Artel (4201541)	0.033790	0.293651	3.782913
Apple (1004227)	LG (1801555)	0.013699	0.333333	2.896825

Based on the research results presented in Table 3, the integration involving Artel (4201541) → LG (1801555) shows a Support value of 0.033790, indicating that this specific integration occurs in approximately 3.38% of all recorded transactions. With a Lift value of 3.782913, this further strengthens the relationship, showing that a customer is nearly four times more likely to interact with LG products after interacting with Artel products, compared to random probability. Additionally, a Confidence value of 0.435294 reflects a conversion success rate of 43.5%, making it one of the most stable behavioral patterns observed in this study.

A similar trend is observed in the rule Apple (1004227) → LG (1801555), which yields a Support value of 0.013699 and a Confidence value of 0.33333. This indicates that one-third of customers interested in Apple products will also be interested in LG devices. With a Lift value of 2.896825, this association demonstrates a strong positive correlation exceeding the threshold for statistical significance. Overall, these findings provide empirical support for the implementation of marketing strategies. A relationship with a high Lift value is highly useful for maximizing

cross-selling on the shopping cart page, optimizing recommendation widgets, and building effective product bundling strategies to increase the overall sales conversion rate.

4.4 Marketing Implications

Based on the results of the comprehensive analysis conducted, the “segment first” strategy emerged as the most optimal marketing strategy among the others. This enables the implementation of more targeted strategies such as personalized promotions and retargeting ads toward visitors who frequently browse the site but rarely take action, such as placing an order.

Clear product relationships can be established through existing frameworks, ranging from automated recommendations and related product ads to package creation. The previously presented dashboard concept remains a crucial tool for tracking changes in data and ensuring marketing strategies align with the latest trends, as consumer habits are rarely permanent.

5. Conclusion

The presented research shows that a systematic application of data mining procedures using the methodology of CRISP-DM can be helpful in finding consumer behaviors patterns in the sphere of e-commerce. The conclusion can be made concerning the significant gap between the stages of interaction and purchasing that proves the existence of the problem associated with shopping cart abandonment. In relation to the research methodology, the combination of such techniques as hierarchical clustering and association rule mining can be seen as much better than those studies that use only one technique at a time. The use of clustering techniques helps find customer clusters and understand their behavioral patterns, while association rule mining can show product-related associations that help explain. consumers' buying behavior. As far as practical applications are concerned, adding visualizations through the use of dashboards could make the results from the analysis easier to use and thus enable converting data analysis into practice. As a result, it would be easier for the company to take an adaptive strategy for marketing with a focus on increasing the conversion rate and target audiences. In spite of the fact that there are 25,000 interactions included in the data set, all of them come from only 100 users. Accordingly, the patterns discovered here should be considered in the light of characteristics of the sample used in this case. The insights obtained from the analysis of the data provided can be very important in terms of understanding customer behavior and their preferences concerning products, but other patterns may appear in relation to significantly bigger samples. Future research might examine the inclusion of real-time analytics and other behavioral indicators that could add to the responsiveness and accuracy of decision making through the model. A more generalized approach involving time-based dynamics and other external factors could be considered.

Data Availability

The dataset analyzed in this study is publicly available from Kaggle (*E-commerce Behavior Data from Multi Category Store*, Mikhail Kechinov) at https://www.kaggle.com/datasets/mkechinov/ecommerce-behavior-data-from-multi-category-store/data. The dataset originates from the REES46 Marketing Platform (https://rees46.com) Open CDP project (https://rees46.com/en/open-cdp) and is used under the attribution requirements specified by the data provider.

Author Contribution Statement

C.B. Joe: Methodology, Software, Validation, Formal analysis, Data Curation, Visualization, Writing - Original Draft, Writing - Review & Editing.

J.C. Purnomo: Methodology, Validation, Formal analysis, Writing - Original Draft, Writing - Review & Editing.

S.A. Triutomo: Conceptualization, Writing - Original Draft, Writing - Review & Editing.

R. Puspita: Writing - Review & Editing, Supervision, Project administration, Funding acquisition.

AI Usage Declaration

In preparing this manuscript, the authors used ChatGPT (OpenAI) to assist with language refinement, grammar correction, and improvement of sentence structure and readability. The AI tool was used solely as a writing support tool and was not involved in the research design, data collection, data analysis, interpretation of results, or generation of scientific conclusions. All outputs generated by the AI tool were carefully reviewed, verified, and edited by the authors. The authors take full responsibility for the content of this manuscript and declare that the AI tool was used responsibly under human supervision.

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Appendix A. Additional Analytical Information

A.1. Clustering and Association Rule Evaluation

Models for customer segmentation were analyzed through Silhouette Score, DBI, and CH Indices. Product association rules, created with Apriori algorithm, were analyzed through the measures of Support, Confidence, and Lift Indices.

A.2. Practical Marketing Applications

The analysis results help with offers, grouping products together for promotions suggesting items to customers and re-engaging them with ads. This can make customers more interested and likely to buy on shopping sites.

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